1. System Overview

Problem

Traditional customer support systems give:

- Generic answers
- Low accuracy
- Poor user experience

Solution

A **RAG-based system** that:

- Reads PDF knowledge base
- Retrieves relevant data
- Generates contextual answers
- Uses workflow + decision logic
- Escalates to human when needed

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2. Architecture Diagram

```
            +----------------------+
            |      User (CLI/Web)  |
            +----------+-----------+
                       |
                       v
            +----------------------+
            |   Query Input Node   |
            +----------+-----------+
                       |
                       v
            +----------------------+
            |  LangGraph Workflow  |
            | (Control Logic)      |
            +----------+-----------+
                       |
                       v
+----------------------------------------+
|          Retrieval Layer               |
| (Convert Query → Embedding → Search)   |
+----------+-----------------------------+
           |
           v
+------------------------------+
|   Vector Database (ChromaDB) |
|   (Stores Embeddings)        |
+----------+-------------------+
           |
           v
+------------------------------+
| Retrieved Relevant Chunks    |
+----------+-------------------+
           |
           v
```

```
        +-----------------------------+
        |   LLM Processing Layer      |
        | (Generate Final Answer)     |
        +----------+------------------+
                   |
         +---------+----------+
         |                    |
         v                    v
+------------------+   +----------------------+
|  Output Response |   |   HITL (Human Help)  |
|  (Answer to User)|   |  Escalation System   |
+------------------+   +----------------------+
```

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3. Document Ingestion Pipeline (Add this mini diagram too)

```
PDF File
   ↓
Document Loader
   ↓
Text Extraction
   ↓
Chunking (300 size, 50 overlap)
   ↓
Embeddings Creation
   ↓
Stored in ChromaDB
```

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4. Component Explanation

User Interface

Takes user query (CLI/Web)

Document Loader

Reads PDF and extracts text

Chunking

Splits text into smaller parts

Embedding Model

Converts text → vectors

Vector Store (ChromaDB)

Stores embeddings for fast search

Retriever

Finds relevant chunks

LLM

Generates answer using context

LangGraph Workflow

Controls flow and logic

Routing Layer

Decides:

- Answer OR escalate

HITL Module

Handles human escalation

5. Data Flow

Full Flow

```
PDF → Load → Chunk → Embed → Store
User Query → Embed → Search → Retrieve → Answer
```